TAMPINES MERIDIAN JUNIOR COLLEGE

JC2 PRELIMINARY EXAMINATION

CANDIDATE NAME	
CIVICS GROUP	

H2 BIOLOGY — 9744/02

Paper 2 Structured Questions

14 September 2022

2 hours

Candidates answer on the Question Paper.
No additional materials are required.

READ THESE INSTRUCTIONS FIRST

Write your name and Civics Group in the spaces at the top of the page.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [ ] at the end of each question or part question.

For examiner's Use	
1	/ 9
2	/ 9
3	/ 12
4	/ 13
5	/ 8
6	/ 8
7	/ 18
8	/ 13
9	/ 5
10	/ 5
Total	**/ 100**

QUESTION 1

Fig. 1.1 is an electron micrograph of a mammalian liver cell undergoing G1 phase of the mitotic cell cycle. The liver cell contains an abundance of mitochondria and some storage molecules.

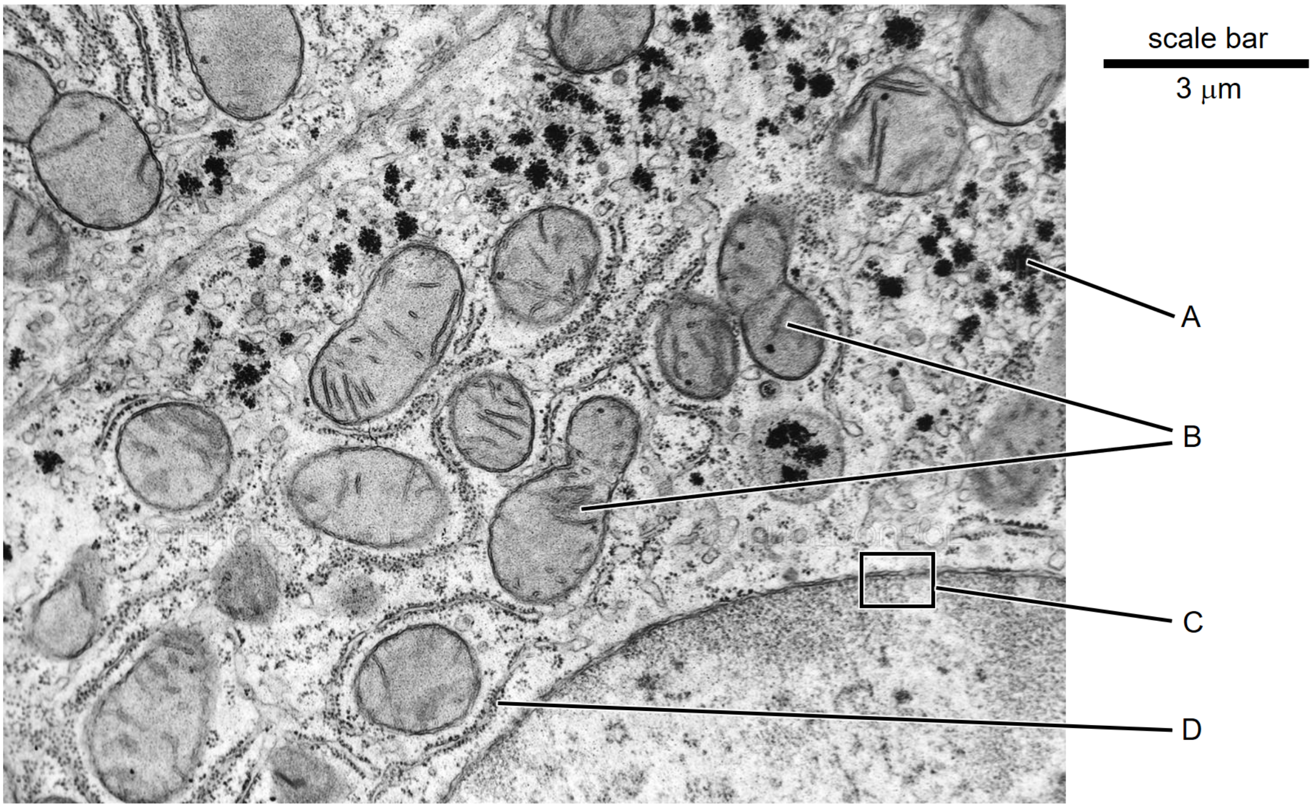

Fig. 1.1

(a) **A** is a storage polysaccharide while **B**, **C** and **D** are organelles or part of an organelle.

Name the storage polysaccharide **A** and organelle **D**. [2]

polysaccharide **A** …………………………………………………………………………………

organelle **D** ………………………………………………………………………………………

(b) Explain why a typical light microscope in a Science laboratory could not have produced the image in Fig. 1.1. [1]

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(c) Calculate the magnification of the electron micrograph. Show your working. [2]

Magnification = ×…………………

(d) The structures labelled **B** on Fig. 1.1 are two mitochondria that are about to divide. Mitochondria synthesize ATP to power cellular processes.

Explain the importance of the division of mitochondria for the cell shown in Fig. 1.1 **and** for subsequent generations of the mammalian liver cells. [2]

for the cell shown in Fig. 1.1

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for subsequent generations of the mammalian liver cells

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(e) Within a cell, substances move between the nucleus and the cytosol. The area labelled **C** in Fig. 1.1 shows an area where this movement occurs.

Fig. 1.2 shows a magnified drawing of area **C**.

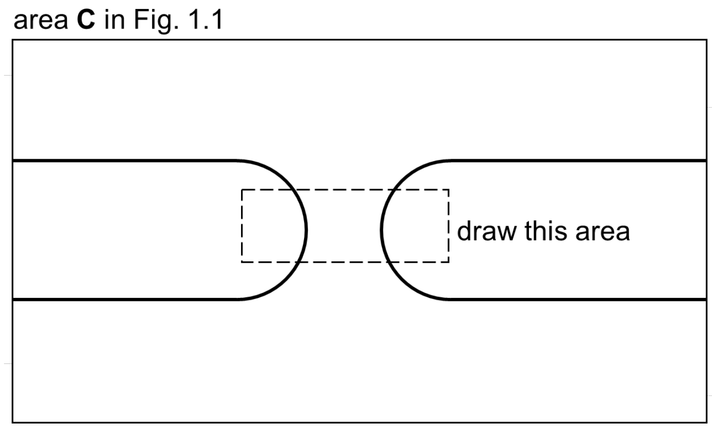

Fig. 1.2

Make a large drawing of the area within the dotted box shown in Fig. 1.2 to show the arrangement of phospholipid molecules. Labeling is **not** required.

You should use the symbol  in your drawing. [2]

[Total: 9]

QUESTION 2

A collagen polypeptide comprises an abundance of the amino acids glycine, proline and hydroxyproline. The repeating tripeptide sequence of glycine–proline–hydroxyproline is very common in collagen.

Fig. 2.1 shows a section of a collagen polypeptide comprising this tripeptide. The R-groups are indicated in bold.

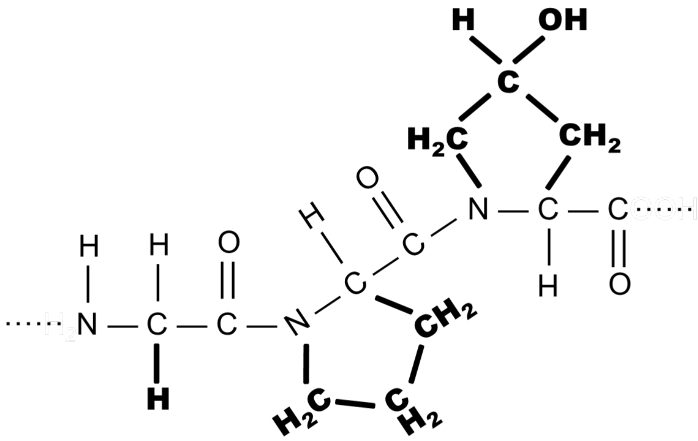

Fig. 2.1

(a) On Fig. 2.1, use arrows to indicate the peptide bonds that join the three amino acids. [1]

(b) Unlike glycine and proline, hydroxyproline is **not** directly added to the collagen polypeptide during translation. After translation is completed, no further amino acid is added.

Explain how a collagen polypeptide eventually contains hydroxyproline. [2]

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(c) Fig. 2.2 shows the structure of a single collagen polypeptide. The R-groups of glycine, proline and hydroxyproline are indicated.

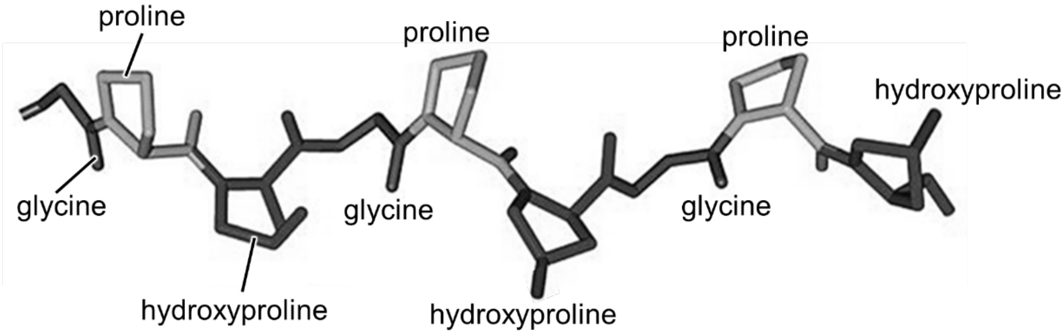

Fig. 2.2

The R-group of glycine is a hydrogen atom. In contrast, proline and hydroxyproline have a rigid and inflexible R-group that tend to cause the polypeptide to bend.

With reference to Fig. 2.2 and the information provided, suggest the role of glycine and that of proline and hydroxyproline in the formation of a tropocollagen molecule. [2]

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proline and hydroxyproline ………………………………………………………………………..

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(d) The enzyme collagenase breaks down collagen. TIMP-1 is a chemical known to inhibit collagenase.

Fig. 2.3 shows the effect of adding a minute amount of TIMP-1 to collagenase. The relative concentration of the products of collagen breakdown was recorded every minute for over seven minutes.

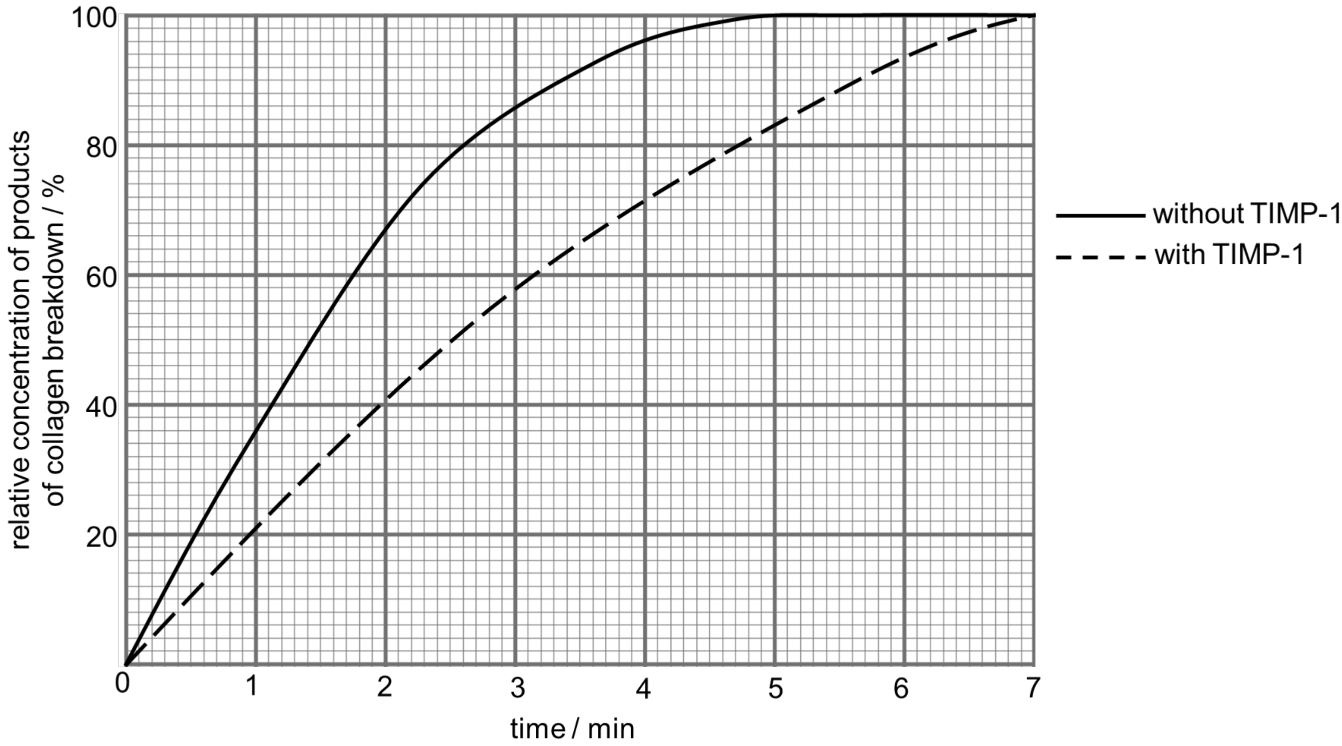

Fig. 2.3

Explain if Fig. 2.3 provides sufficient evidence to conclude if TIMP-1 is a competitive or non-competitive inhibitor of collagenase. [4]

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[Total: 9]

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QUESTION 3

Fig. 3.1 represents some of the reactions that take place in a leaf cell of a flowering plant.

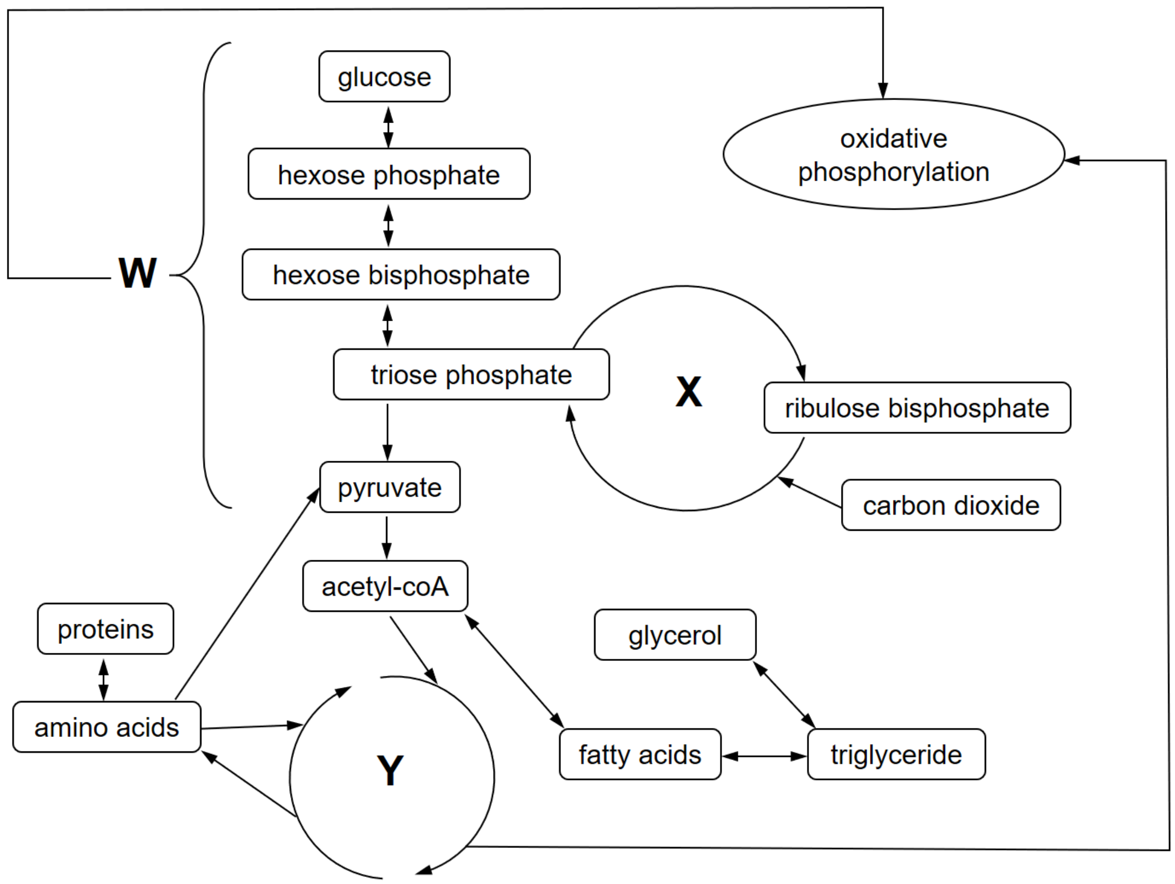

Fig. 3.1

(a) Name the reaction pathways indicated by the letters **W**, **X** and **Y**. [3]

W ....

X ....

Y ....

(b) Explain how the three reaction pathways (**W**, **X** and **Y**) are able to work independently of one another in the same leaf cell. [2]

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(c) With reference to the metabolic reactions outlined in Fig. 3.1, state the roles of **three named** coenzymes in this leaf cell. [3]

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(d) The carbons in glucose and the carbons in the fatty acids of triglyceride are used to generate acetyl-coA, a two-carbon molecule.

Using your knowledge on biomolecules and the information in Fig. 3.1, explain why **one molecule** of triglyceride can generate more ATP than **one molecule** of glucose. [4]

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QUESTION 4

A number of different proteins are involved in regulating the cell cycle.

To initiate mitosis, cyclin B and CDK1 (a protein kinase) bind to each other to form a complex called the mitosis-promoting factor (MPF). CDK1, when unbound to cyclin B, is inactive.

MPF phosphorylates proteins that are important for the cell to enter prophase. Examples of these proteins are:

- condensins – the **phosphorylated** form binds to chromatin threads to promote condensation into chromosomes
- lamins – the **unphosphorylated** form binds complementarily to the underside of the nuclear membrane to provide structural support to the nucleus

(a) Explain how phosphorylation of **lamins** by MPF allows the cell to **complete prophase**. [2]

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The condensed chromosomes visible in prophase each consists of sister chromatids, which are held together along their length, including the centromere, by proteins known as cohesin.

Cohesin can be broken down by the enzyme separase.

Separase is initially synthesised in an inactive form by binding to another protein called securin, which physically blocks the active site of separase.

(b) Draw a **labelled** diagram to show how sister chromatids are held together. [3]

(c) Later in mitosis, MPF activates a protein complex called the anaphase-promoting complex (APC).

One role of the activated APC is to catalyse the transfer of ubiquitin molecules to securin and cyclin B. Ubiquitinated proteins are bound for degradation in the proteasome.

Using **all** the information given in this question thus far, explain how

(i) degradation of securin triggers the cell to enter anaphase. [3]

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(ii) degradation of cyclin B triggers the cell to enter telophase. [3]

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(d) Unlike mitosis, meiosis gives rise to genetically different cells.

Outline **two** ways by which meiosis can give rise to genetically different cells. [2]

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2. ……………………………………………………………………………………………………..

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QUESTION 5

DNA methylation usually occurs at the cytosine bases of eukaryotic DNA, catalyzed by DNA methyltransferases (DMT). The methylated cytosine residues are usually immediately adjacent to a guanine nucleotide, resulting in two methylated cytosine residues sitting diagonally to each other on opposing DNA strands.

Different variants of the DMT act in different ways. Fig. 5.1 shows the role of two different variants of DMT (DMT3 and DMT1) at different junctures of DNA methylation.

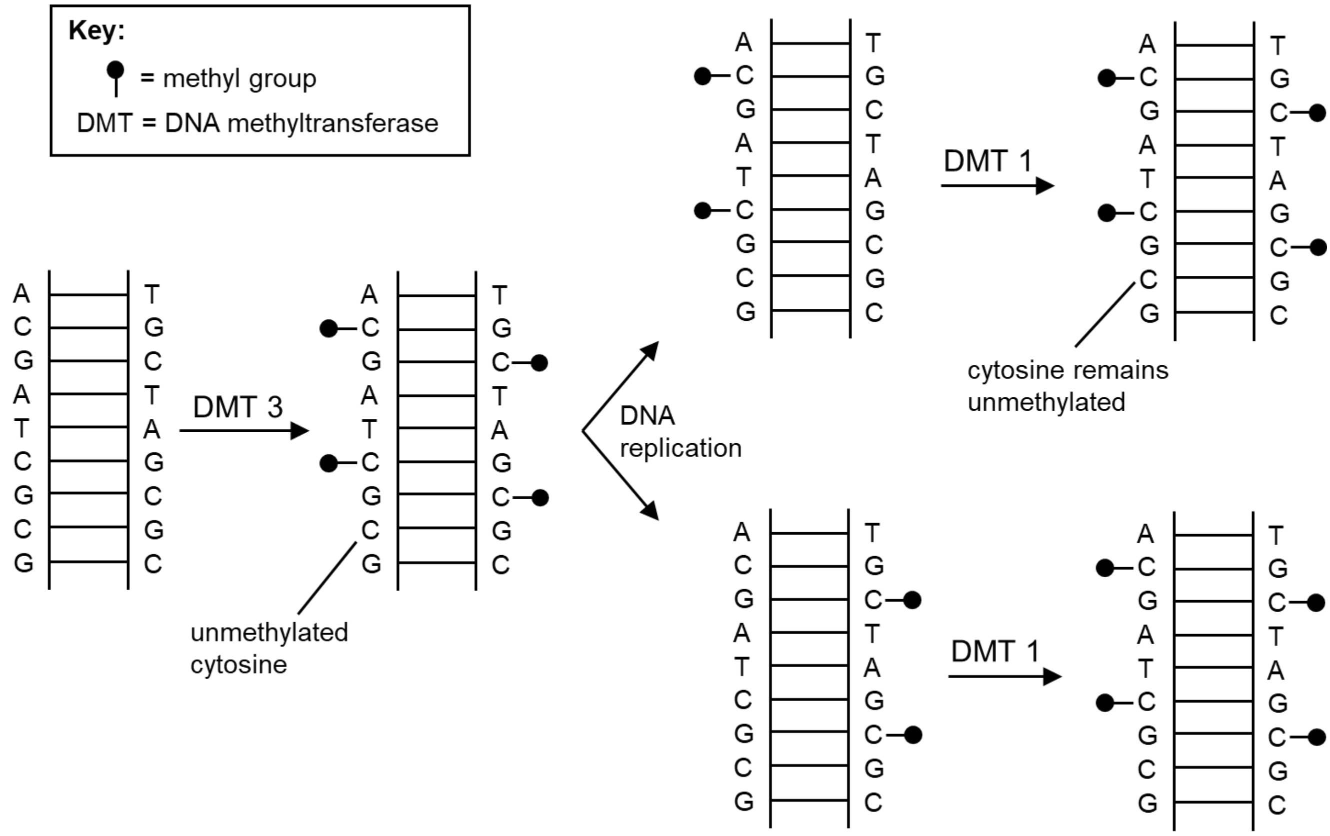

Fig. 5.1

(a) State the name of the enzyme that uses the base sequence of one DNA strand to form the complementary strand. [1]

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(b) With reference to Fig. 5.1, state the role of DMT3 and DMT1 in DNA methylation. [2]

DMT 3 ....

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DMT 1 ....

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The *APC* gene was implicated as a possible cause of colon cancer. Samples taken from colon cancer cells of 10 patients and colon cells of 10 healthy individuals were analyzed for their methylation patterns.

Fig. 5.2 shows the DNA methylation patterns on the promoter of the *APC* gene from the 20 samples. The numbers (32–269) represent the base position on the promoter.

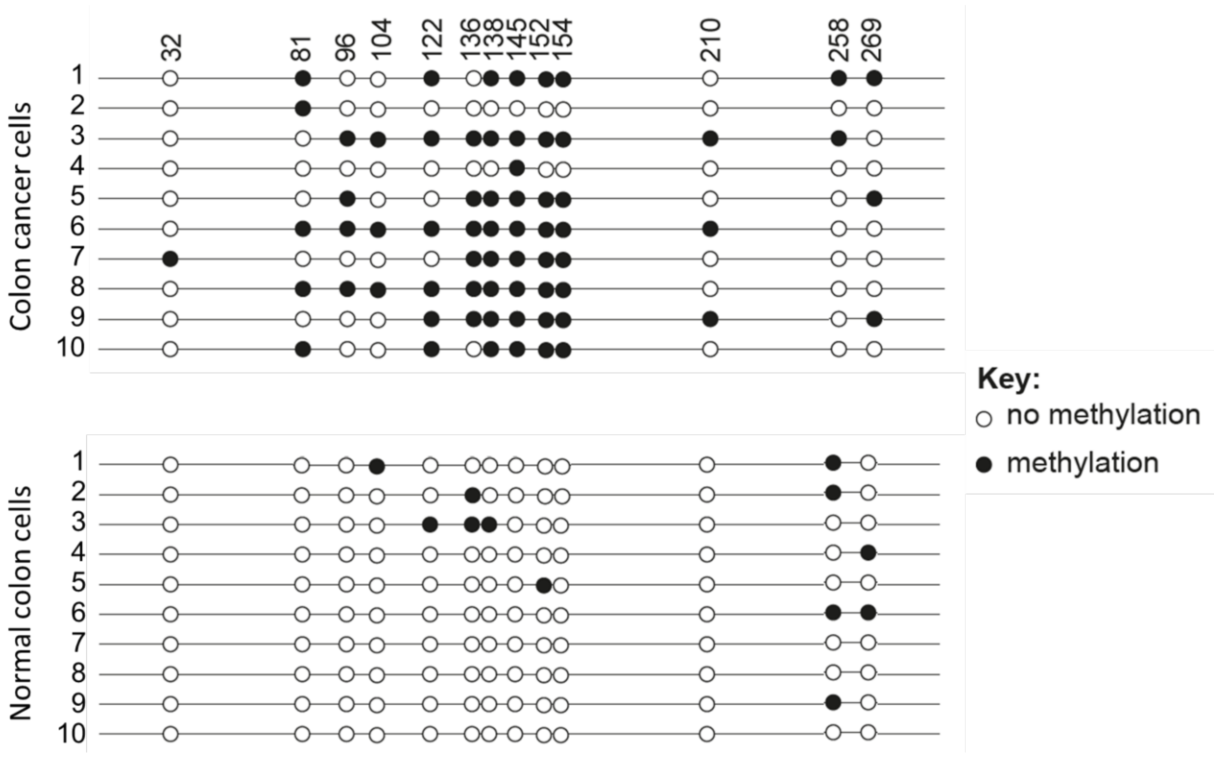

Fig. 5.2

(c) Deduce if the *APC* gene is a proto-oncogene or a tumour-suppressor gene. [1]

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(d) Contrast the methylation patterns of cancer and normal samples. [2]

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(e) Suggest why some healthy individuals did not develop colon cancer, despite the *APC* gene being methylated. [2]

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[Total: 8]

QUESTION 6

In white clover, *Trifolium repens*, one gene determines the production of a cyanide-forming substrate. Allele **A** produces the cyanide-forming substrate, whilst allele **a** produces no substrate.

A second gene, located on a different chromosome, determines the production of an enzyme that catalyses the release of cyanide from the substrate. Allele **E** produces the enzyme, whilst allele **e** produces no enzyme.

- Clover that has both **A** and **E** alleles releases cyanide rapidly when its leaves are crushed.
- Clover with **A** but not **E** releases cyanide slowly when its leaves are crushed.
- Clover that does not have **A** cannot release cyanide.

In an experiment, a clover that releases cyanide rapidly was self-pollinated. The following numbers of offspring were obtained:

rapid cyanide release	140
slow cyanide release	49
no cyanide release	67

(a) Construct a genetic diagram to explain the above offspring numbers. [4]

(b) A student attempted to perform a chi-squared (χ^2) test on the offspring numbers.

(i) Using the ratio from part **(a)**, complete the table below to show the **expected** number of each phenotype of the offspring. [1]

phenotype	observed number (O)	expected number (E)
rapid release	140	…………
slow release	49	…………
no release	67	…………

(ii) Table 6.1 shows part of the χ^2 table.

Table 6.1

Degrees of freedom	Probability								
	0.9	0.8	0.7	0.5	0.2	0.1	0.05	0.02	0.01
1	0.016	0.064	0.15	0.46	1.64	2.71	3.84	5.41	6.64
2	0.21	0.45	0.71	1.39	3.22	4.60	5.99	7.82	9.21
3	0.58	1.00	1.42	2.37	4.64	6.25	7.82	9.84	11.34
4	1.06	1.65	2.20	3.36	5.99	7.78	9.49	11.67	13.28

The calculated χ^2 value is determined to be 0.273.

Explain what can be concluded from the calculated χ^2 value of 0.273. [3]

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[Total: 8]

QUESTION 7

Erythropoietin (EPO) is a glycoprotein hormone produced by the kidney in response to low blood oxygen concentration. EPO binds to its target stem cell to promote the formation of red blood cells.

(a) State the name of the stem cell to which EPO binds **and** describe the unique features of this stem cell. [3]

name ....

features ....

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(b) Fig. 7.1 shows the cascade of events that occur upon binding of EPO to the EPO receptor on the target stem cell, leading to the activation of the transcriptional activator, GATA1.

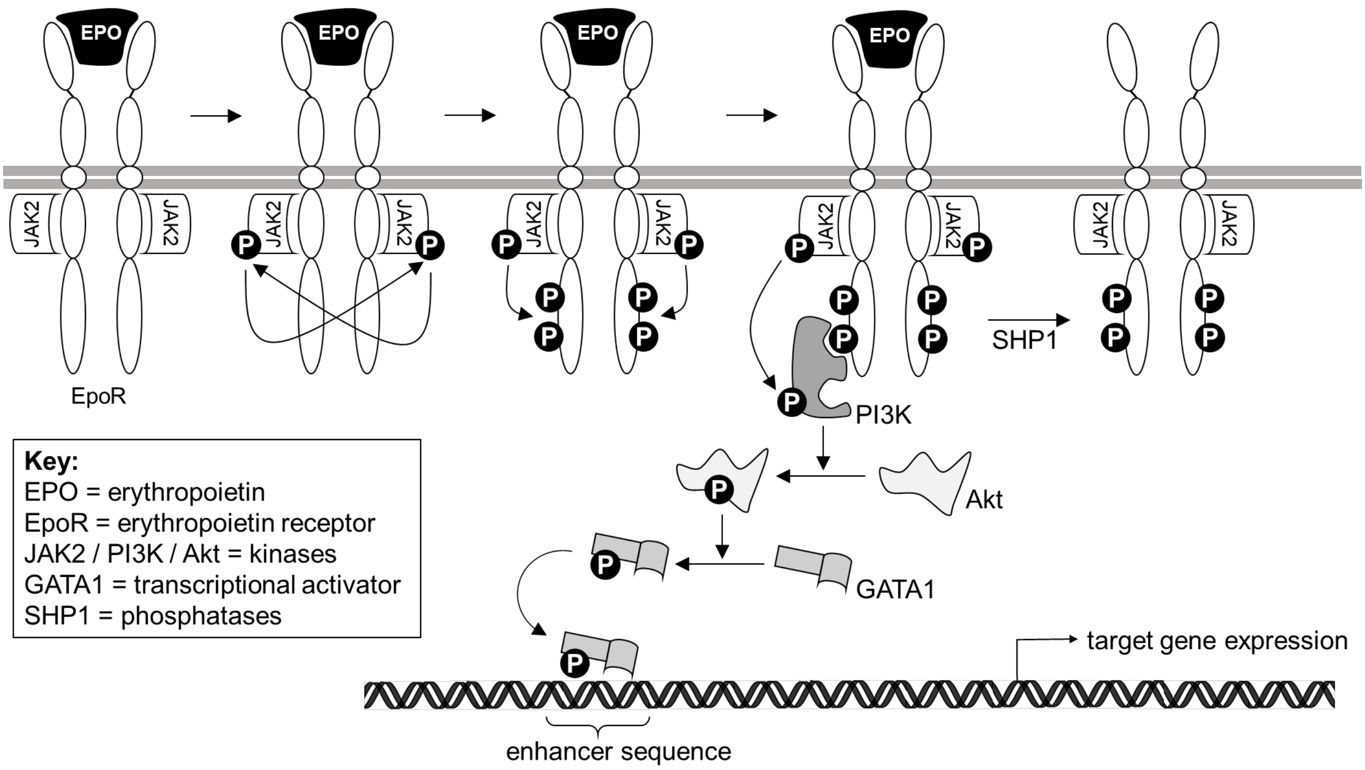

Fig. 7.1

(i) Describe the signal **transduction** events leading to the activation of the transcriptional activator, GATA1. [4]

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(ii) Explain how the transcriptional activator GATA1 **upregulates** the expression of target genes. [3]

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(iii) Name one possible target gene that can be upregulated by GATA1. [1]

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(iv) Explain how EPO signaling is terminated. [1]

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(c) As developing red blood cells mature during blood synthesis, membranous organelles such as mitochondria are lost from the cell to make space for haemoglobin molecules. Many organelle-dependent reactions hence do not occur in red blood cells.

(i) Suggest how the loss of mitochondria is a **further** advantage to the proper functioning of red blood cells. [2]

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(ii) Explain how red blood cells are still able to synthesize a small amount of ATP throughout their lifespan, despite having lost their mitochondria. [2]

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(iii) Despite the lack of genes due to the absence of a nucleus, new protein molecules continue to be synthesised throughout the lifespan of a red blood cell. However, the rate of protein synthesis decreases with the lifespan of the red blood cell and eventually ceases.

Explain why new proteins can still be synthesised by a red blood cell **and** why protein synthesis eventually ceases. [2]

why new proteins can still be synthesised …………………………………………………………

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why protein synthesis eventually ceases …………………………………………………………..

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[Total: 18]

QUESTION 8

Science has identified some two million species of plants, animals and micro-organisms on Earth, but scientists estimate that there are thirteen million more left to discover. The most commonly discovered new species are typically insects, a type of animal with a high degree of biodiversity and accounts for more than half of the species identified. New species are typically discovered in remote places that have not been well studied previously, such as islands.

(a) Suggest why it is impossible to identify all living species on earth. [1]

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(b) Suggest **and** explain why there are so many different species of insects as compared to other animals. [2]

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(c) Scientists have long been puzzled over how land animals reach remote islands that lie in the midst of vast oceans. Recent studies suggest that animals can travel from one land mass to another by floating on giant rafts of earth and vegetation.

(i) Suggest how climate change may facilitate the colonization of remote islands by land animals. [1]

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(ii) Explain why species on remote islands are usually not found anywhere else on earth. [3]

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(iii) Explain why island populations are much more prone to extinction than mainland populations. [1]

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(d) Cryptic species are animals that look outwardly similar to another species. Oak Titmouse and Juniper Titmouse, both native to North America, are examples of cryptic species. Until recently, they had been considered as the same species for 151 years.

Fig. 8.1 shows the titmice and their distribution on the North America continent.

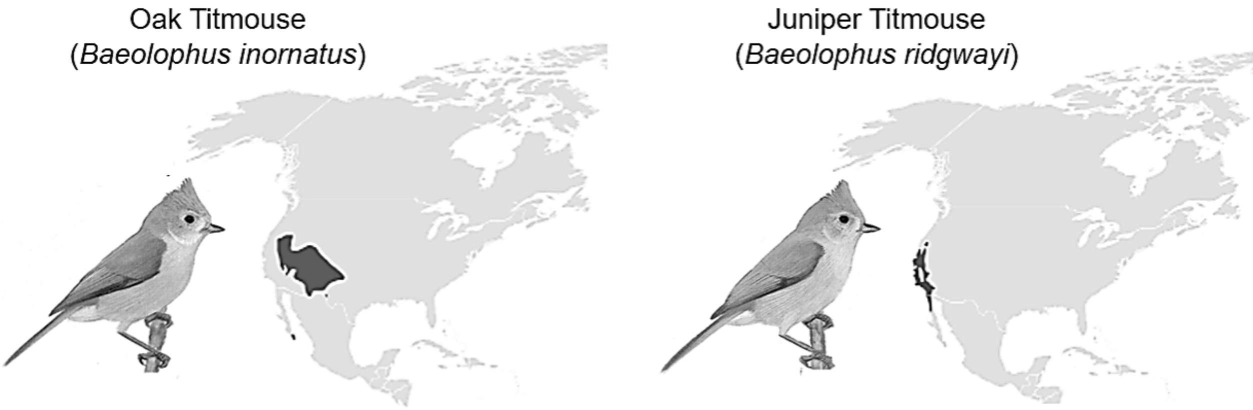

Fig. 8.1

(i) State the **lowest** taxonomic rank on the Linnaean system of classification to which the two populations of titmice share. [1]

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(ii) Explain why Oak Titmouse and Juniper Titmouse are now classified as two different species. [2]

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(iii) Suggest **and** explain a **pre-zygotic** isolating mechanism that could prevent successful reproduction between the two populations of titmice in captivity. [2]

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[Total: 13]

QUESTION 9

In mammals, three genes code for the production of antibodies. Fig. 9.1 shows the structural arrangement of the three genes in humans.

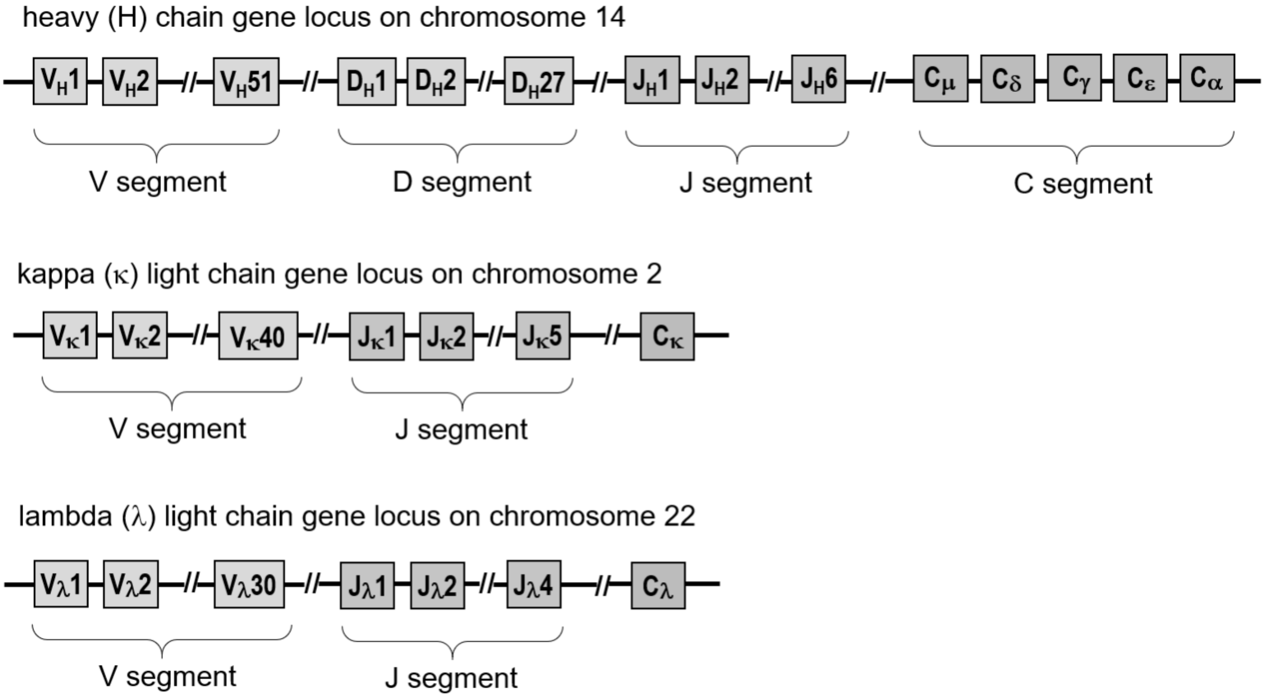

Fig. 9.1

Explain how the structural arrangement of the antibody gene loci creates millions of different antibodies that vary at their antigen-binding site. [5]

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[Total: 5]

QUESTION 10
The deer tick (*Ixodes scapularis*) is an arthropod which sucks blood from humans and other mammals. Some deer ticks can be infected by the bacterium *Borrelia burgdorferi*. When a tick bites a human, the bacterium is often introduced, causing Lyme disease. Lyme disease is a public health problem in North America and, if left untreated, can cause neurological impairment.

Fig. 10.1 represents the two-year life cycle of a tick.

The range of hosts at the three key developmental stages of its life cycle are:

- larva – birds and small mammals
- nymph – humans and other small to large mammals
- adult – humans and other large mammals

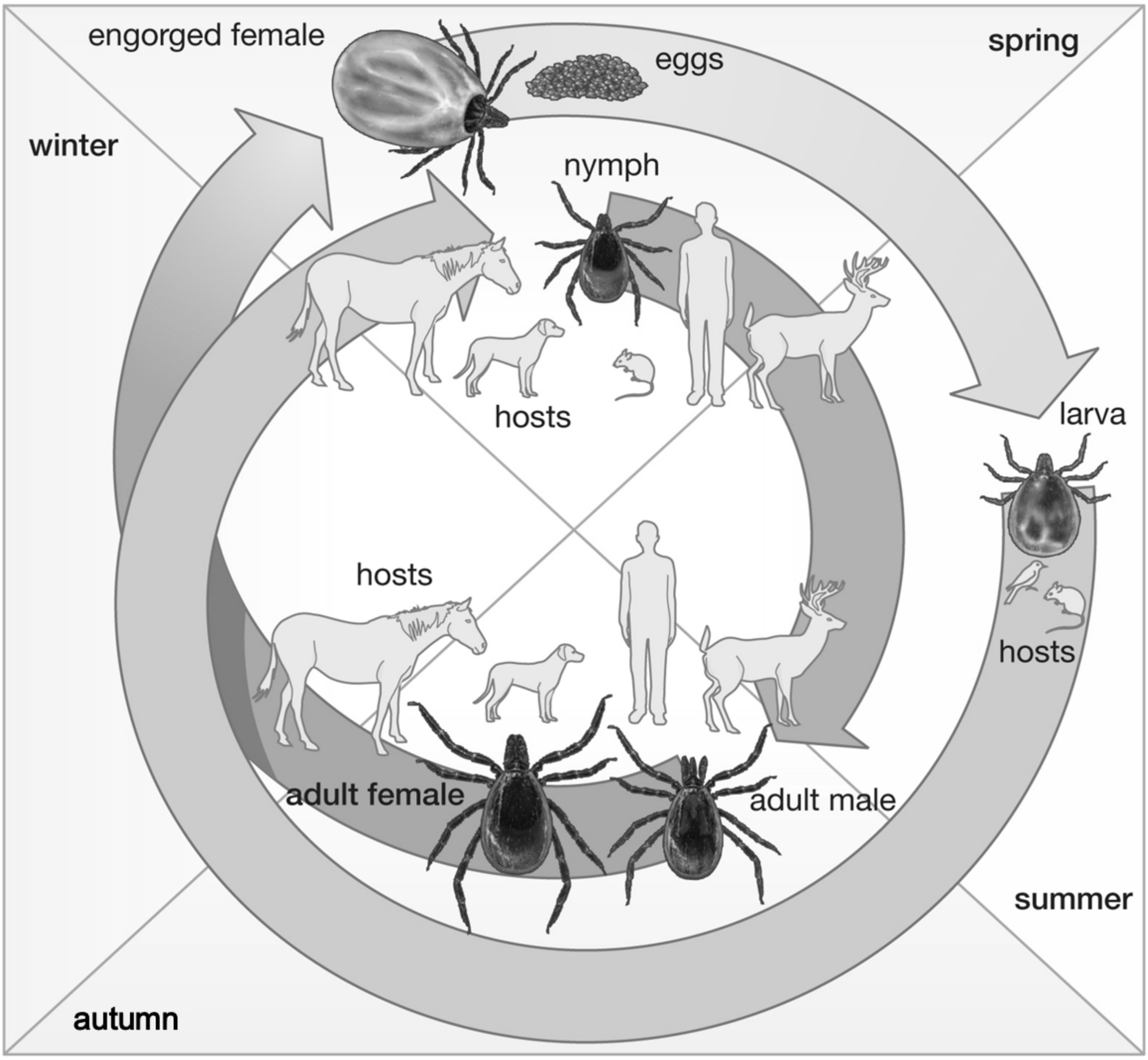

Fig. 10.1

(a) Using the information provided, suggest one possible treatment for Lyme disease. [1]

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Fig. 10.2 shows the developmental stages of deer ticks throughout the four seasons in a densely human-populated area of Canada for the year 2000 and predicted for the year 2080 based on the rate at which the earth is warming.

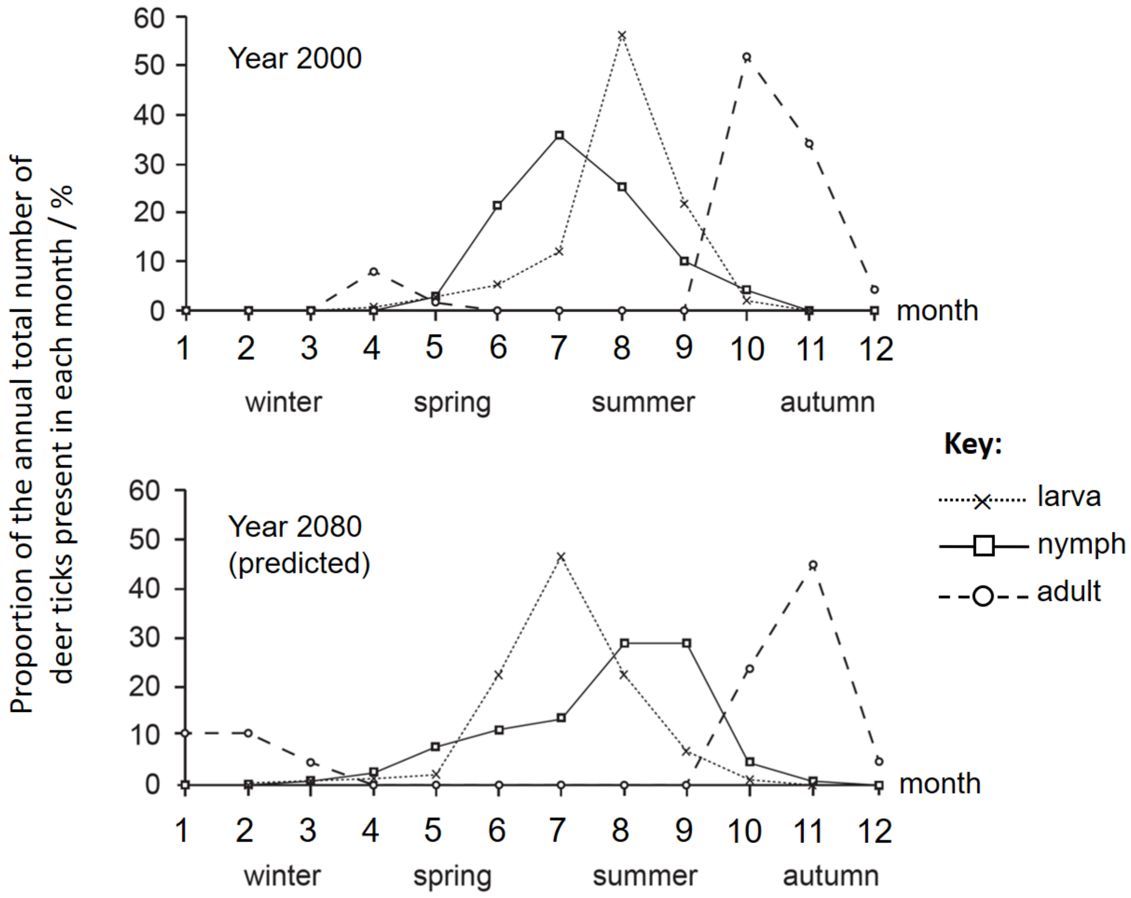

Fig. 10.2

(b) Using Fig. 10.1 and Fig. 10.2, evaluate how global warming will affect the spread of Lyme disease in 2080. [4]

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[Total: 5]

☺ **END OF PAPER 2** ☺

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